HealthCare Domain Testing with Sample Test Cases

Before we begin testing, let's quickly study the basic healthcare domain knowledge.

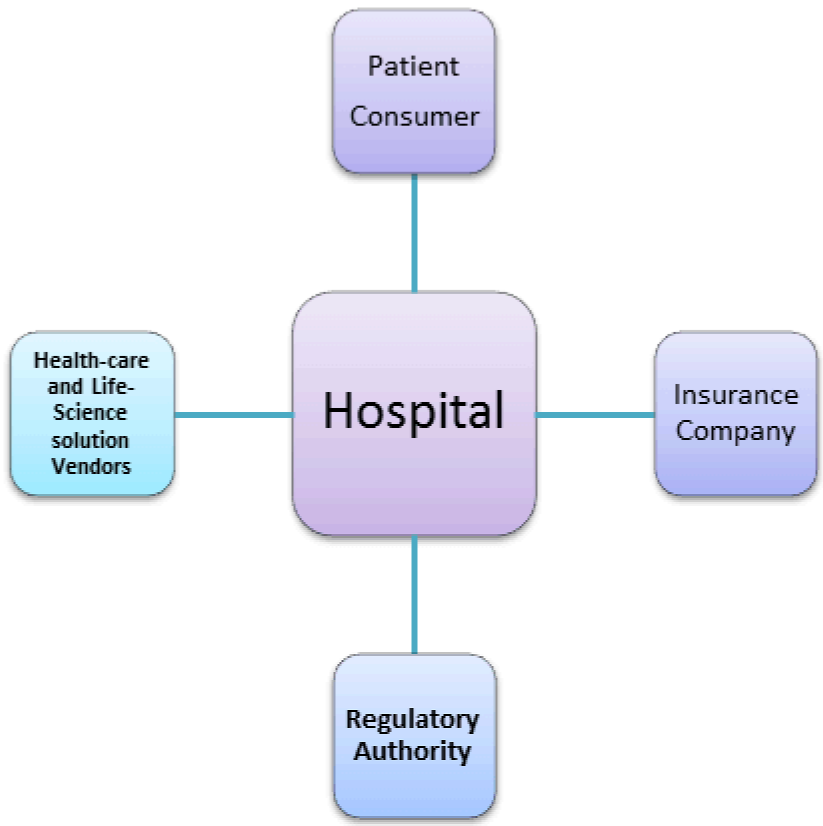
HealthCare Domain Testing

Healthcare Domain Testing is a process to test healthcare application for various factors like standards, safety, compliance, cross dependency with other entities, etc. The purpose of healthcare domain testing is to ensure quality, reliability, performance, safety and efficiency of the Healthcare application.

Basic knowledge of Health Care Domain

The entire health care system is weaved with each other by the single body that is hospital or provider (doctor).
While the other entities include-

- **Insurance company:** Medicare, Medicaid, BCBS, etc.
- **Patient/Consumers:** Patient Enrolled
- **Regulatory Authority:** HIPAA, OASIS assessment, HCFA 1500 and UB92, etc.
- **Health-care and Life-Science solution Vendors**



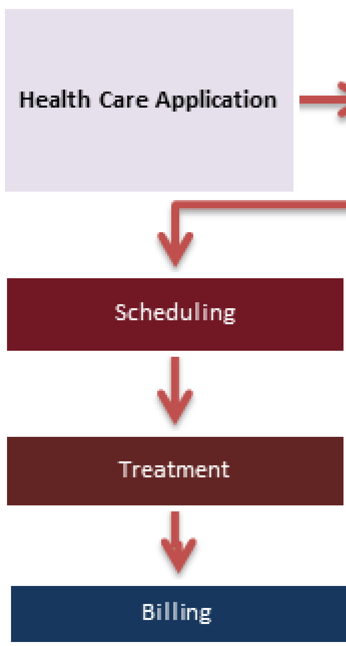
Basic Terminology of Health Care System

- **Provider**: A health care professional (doctor), medical group, clinic, lab, hospital, etc. licensed by health care services
- **Claim:** A request to your health insurance company to pay a bill for health care service
- **Broker:** An insurance professional, who negotiates, procures insurance on behalf of insured or prospective insured
- **Finance:** Insurance bodies that pay for medical expenses, it could be government (Medicare or Medicaid) or commercial (BCBS)
- **Medicare:** A federal health insurance program for senior citizen and permanently disabled people
- **Medicaid:** A joint and state program that helps low-income families and individuals pay for the cost associated with medical care

- **CPT code**: A current procedural terminology code is a medical code set to describe medical, surgical and diagnostic services
- **HIPAA**: It is a set of rules and regulations which doctors, hospitals, healthcare providers and health plan must follow in order to provide their services

Healthcare Business Process

Most health-care organization have adapted software program to process the smooth functioning of the system. This software system gives all the information in a single document for each entity dealing with this.



Interconnecting this whole system to a single web application is a huge task and making it work effectively is even a bigger task. Rigorous testing of this health application is compulsory, and it has to go through various testing phases.
In this tutorial, we will learn,

Testing of Providers system

Sample Test Scenarios and Test cases for providers (doctor/hospital) system:

Sr #	Test Scenario	Test Cases
1)	Access to providers system	• Provider system should let us enter, edit and save the provider's data
2)	Positive flow System Testing	• It includes scenarios to enter different types of provider, change providers details, save and inquire them
3)	Negative flow System Testing	• Allows to save provider information with incomplete data, contract's effective date, entering details about existing providers in the system
4)	System Integration Testing	• Validate the feed to members system, finance system, claim system, and provider portal. Also, validate if the changes from provider portal are entered into the respective provider's record
5)	Positive flow providers portal testing	• Login and view providers details, claim status, and member details • Make change request to change the name, address, phone number, etc.
6)	Negative flow providers portal testing	• View the member details with an invalid ID

		• Login with invalid credentials
7)	Positive flow Broker portal testing	• Login and view details about broker and commission payment • Make a request to change the name, address, phone number, etc.
8)	Negative flow Broker portal testing	• It should include scenarios to log in with invalid credentials

Testing of Broker System

Sample Test Scenarios and Test cases for Broker System:

Sr #	Test Scenario	Test Cases
1)	Broker System	• It should be capable of edit, enter and save broker data • Broker commission calculation based on the premium payment details from the member system
2)	Positive Flow System Testing	• Enter, save and edit brokers record for different types of broker • For active brokers calculate the commission by creating a feed file with the respective record for members with a different plan

3)	Negative flow System Testing	● Enter a broker record with incomplete data and save for different types of broker ● By creating the feed file with the respective record for members with different plan calculate the commission for the terminated broker ● By creating the feed file with the respective record for members with different plan calculate the commission for the invalid broker
4)	System Testing	● To downstream system such as finance system, broker portal and member system validate the feeds ● Validate if the changes from broker portal are incorporated in the respective broker record

Testing of Member System

Sample Test Scenarios and Test cases for Member (Patient) System:

Sr #	Test Scenario	Test Cases
1)	Member system	● Enroll, reinstate and terminate a member ● Remove and add a dependent ● Generate premium bill ● Process premium payments

2)	Positive Flow System Testing	- With the current, past, and future effective dates enroll different types of members - Inquire and change members - Produce premium bill for an active member for the following month - Terminate an active member with past, current and future termination dates greater than the effective date - Re-enroll a terminated member with current, past and future effective dates - Reinstate a terminated number
3)	Negative flow System Testing	- With insufficient data enroll a member - For a terminated member produce a premium bill for the following month
4)	System Integration Testing	- Validate the feed to downstream systems such as provider portal, broker portal, finance system, and claim system - Validate if the alterations from member portal are incorporated in the respective member record - Process the payment of premium bill generated with the feed from members portal that has details of payment made

Testing of Claims System

Sample Test Scenarios and Test cases for Claims System:

Sr #	Test Scenarios	Test Cases

1)	Claim System	● Claims in health-care should edit, enter and process claims for a member as well as dependent ● For invalid claims, it should throw errors when incorrect data is entered
2)	Positive Flow System Testing	It should include the scenario to edit, enter and process claims for a member as well as dependent
3)	Negative Flow System Testing	● It should validate and enter a claim with invalid procedure code and diagnosis code ● Validate and enter a claim with the inactive provider ID ● Validate and enter a claim with a terminated member
4)	System Integration	It should include a scenario to validate the feed to downstream systems such as provider and finance portal

Testing of Finance System

Sample Test Scenarios and Test cases for Finance System

Sr #	Test Scenarios	Test Cases
1)	Finance System	Enroll, reinstate and terminate a member

2)	Positive flow system testing	It should check whether correct account number or address is chosen for the respective member, provider or broker for the payment
3)	Negative flow system testing	• Verify whether payment is done for an invalid member, provider or broker ID by creating a respective record in the feed • Verify whether payment is done for an invalid amount for the member, provider or broker by creating respective records in the feed

Testing for regulatory compliance

Protecting patient sensitive data and health information is an utmost priority for health regulatory bodies. The testing should be done in compliance of such regulatory bodies.

Sample Test Scenarios and Test cases for Regulatory Compliance:

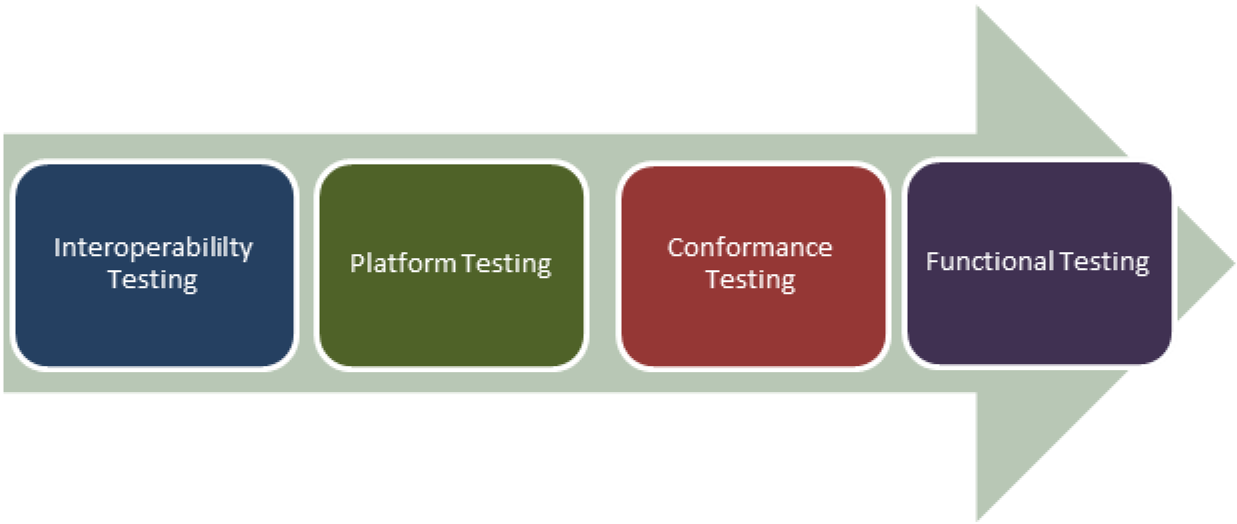
Sr #	Test Scenarios	Test Cases
1)	User's Authentication	Using verification method to ensure that correct users get a login and deny to others
2)	Information Disclosure	Authorizing access to information is based on the user's role and patient limitation
3)	Data Transfer	At all transfer, points ensure that data is encrypted

4)	Audit Trail	All transactions and all attempts to access data with a proper set of audit trail information are recorded
5)	Sanity Testing related to regulatory body	Perform sanity testing and verify the encryption of the data is done in particular areas like EPHI (Electronic Protected Health Information)

Performance testing of Healthcare Application

Before preparing test scenarios certain requirement of the system should be considered. For example, health-care providers (Doctors/Hospitals) provide care 24/7, so the patient check-in software needs to be available at all times. Also, it needs to communicate with insurance companies to validate policy information, send claims and receive remittances. Here, the architecture should define the different components of the system, the protocol to communicate with insurance companies, and how to deploy the system so that it complies 24/7.
As a tester, you need to ensure that the healthcare software system meets the desired load/performance benchmark.

Other Testing Types for Healthcare Application



- **Functional Testing**: Testing healthcare application against functional capabilities
- **Conformance Testing**: Conformance test Healthcare security requisites and industry frameworks
- **Platform Testing**: Testing of applications on Mobile platform and applications testing for cross-browser compatibility
- **Interoperability Testing**: Testing conformance to interoperability standards (Eg; DICOM, HL7, CCD/CDA)

Testing Challenges in Healthcare Application

Testing challenges in testing healthcare application are no different than other web application testing.

- Requires expertise in testing, and usually, it is high in cost
- Requires interoperability, compliance, regulatory, security, safety testing besides regular testing techniques (Non-Functional, Functional and Integration testing)
- Testing should be done keeping in mind the safety and regulatory standards- as any error can cause a direct effect on patient's life
- Testing team needs to be well aware of the various functionalities, clinical usage, and the environment the software will be used for

- A health-care product should comply with various standards like FDA, ISO, and CMMI before it can be used
- Cross dependency of software- testers need to ensure that any changes in one component or layer should not lead to side effect on the other.

Healthcare device Testing

While health-care device software is not the direct concern of patient, they also require rigorous testing like another software testing. For example, X-ray machines that are controlled by software programs should be tested well because any testing error in software can lead to a serious effect on the patient.
FDA (Food and Drug Administration) has guidelines for mobile and web applications for medical devices. While testing medical devices the proper functional Test Plan along with pass and fail criteria is also the part of FDA guidelines. When a test plan is executed, the results are collected and reported to FDA. This process ensures that the device meets the standard of the regulatory bodies.

Useful tips for Healthcare Testing

While testing software, you can consider some important tips for the testing healthcare system.

- Dates are important and need to be accurate
- While designing test cases consider various parameters like different types of plan, brokers, members, commission, etc.
- Complete knowledge of the domain is required